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CONTROLLING AND MONITORING SEMICONDUCTOR MANUFACTURING PROCESS USING SMART TECHNOLOGIES

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ABSTRACT

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Manufacturing semiconductor is one of the most important and technologically advanced industrial sectors.

Controlling and monitoring the semiconductor manufacturing process drastically impact overall production performance because it reduces the gap between the manufacturing process and the supply chain. Smart technologies and methodologies have proliferated in the semiconductor manufacturing process in order to improve productivity and decrease maintenance costs.

The immense, time-sensitive, and diversified nature of the data brings significant challenges to the real-time collection, processing, and decision making.

Purpose of the paper:

The main goal of this work is to provide information about the usage of statistical modeling, IoT, sensors, big data, and control systems used in semiconductor manufacturing with the intention of improving production efficiency, control process defects, and reducing machine downtime. Finally, presented a framework of an industrial data management system that is able to process huge industrial data, support monitoring, and control in the manufacturing process.

I. INTRODUCTION

The semiconductor played a huge role in the history of mankind because of its effect on accelerating technology and advancing the world's economy in the 20th century. Gian et al. (2012) in their work emphasize and addressed issues and challenges in automatic Control and Machine Learning for Semiconductor Manufacturing: Review and Challenges [[2]]. When the semiconductor was born in the Bell Labs in 1948, its effects are largely unexpected. A megabit of memory cost at that time is $75 Million, but now it is worth around 15 cents. Semiconductors together with VLSI and ULSI brought huge changes in the electronic industry so far.

Because of the increasing ability to store and process information, it has become an extremely important operator of social, scientific, economic, and legal processes. Semiconductors play a major role in every electronic device, for example, personal computers, mobile phones, refrigerators, and so on.

Most of the semiconductor manufacturing companies give high priority to improve quality and computation capability in the manufacturing process. Semiconductor companies are investing a lot of resources because it would be possible to improve manufacturing capabilities using statistical prediction and control systems to decrease defective products and increase process stability. Transmitting real-time data across the industrial network IoT creating manufacturing intelligence.

There are a lot of observable parameters such as historical time series, sensor data, a characteristic behavior of the equipment, using all of these parameters possible to predict the future status of the equipment, and making strategic decisions about maintenance and production.

Produced an immense amount of industrial data used for prognosis, prediction, real-time control, and monitoring manufacturing activities. This data helps organizations for failure identification and prevention, predicting machine health status, and making smarter business decisions.

The future of electronic manufacturing industry depends on smart manufacturing because it improves production efficiency, safety and security using emerging technologies such as industry4.0, IoT, effective

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sensors, and artificial intelligence enables real-time analytics for the predictive process; machine control and visualization in the manufacturing process.

II. RELATED WORK

Smart manufacturing increases economy, safety, gives complete control on manufacturing process; supports reducing global warming; suitable for rapid change in the complex business process; handling increasing factory intricateness; intended to streamline reliability, productivity, capability, flexibility and cost requirements in the manufacturing process; chain with global market demands using advanced technologies like industrial internet of things, big data solutions, industry 4.0 and artificial intelligence Methodologies.

Many efforts have been made for the IoT data acquisition, preparation, processing, and storage in smart manufacturing. Industrial internet of things connects billions of devices intended to revolutionize the manufacturing industry, increase process efficiency, reliability, and safety. Smart manufacturing increases product quality, time to market, increases customer satisfaction and enhances product market size.

In the worldwide manufacturing process, industrial data [[27]] plays a crucial role intended to generate huge wealth and create new job opportunities in developing as well as developed countries. Industrial data are complex and unstructured, managing and storing this data requires [[37]] high processing power. IoT, sensors, Industry 4.0, and predictive models help to build an efficient production system in smart manufacturing.

Manufacturing plays an important role in developing human living standards, and it is a crucial part of

economic growth in every sector in this globalization era.

Emerging technologies are used by the manufacturing industry to magnify the economic competitiveness of individual manufacturers and the sustainability of the entire industrial sector using smart technologies as shown in figure 1.

Fig-1: Using smart technologies in semiconductor manufacturing

Emerging technologies bring significant improvements in the manufacturing process and influence reducing energy consumption, improving production ability, diminishing social risks, and influencing decision making.

Big data technologies reduce the processing time, provides effective solutions from generating huge data volumes in real-time. Technologies like Apache Kafka, apache storm and MongoDB add significant benefits to big data processing and storage in semiconductor manufacturing.

IoT based sensors add significant advantages to collect data for big data processing.

Generated data from manufacturing industries were analyzed using machine learning (example fault detection, quality prediction, defect classification, and run to run control) technologies to make suitable decisions in the

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company. The growth of big data in recent trends reduces the gaps in technology and contributes to new technological trends for the future.

Higher volumes of data generation will impact positively on developing more robust real-time solutions in the industry. In semiconductors, the connecting architecture enhances the smart industry, increases communication between devices, and real-time monitoring and controlling.

This paper is organized as:

Section 3. Semiconductor device fabrication process.

Section 4. Using smart technologies in the semiconductor manufacturing process.

Section 5. Presented an "Architecture for monitoring and controlling the end to end semiconductor manufacturing process in industries." that utilizes the above-mentioned technologies such as IoT based sensors, big data, and machine learning models to control and monitor semiconductor manufacturing.

Section 6. Challenges and future work are presented.

III. SEMICONDUCTOR DEVICE FABRICATION PROCESS

The process used to manufacture semiconductor devices is called Semiconductor device fabrication as shown in figure 2. Usually, the semiconductor manufacturing process takes 8 weeks of time. In the manufacturing process, integrated circuits with components such as transistors are formed on the surface of a silicon crystal wafer.

In integrated circuit formation, a thin film layer will form the wiring, the photoresist is coated with this thin film; as a mask for etching to process the thin film into the shape of the wiring and other components, this photoresist is used to shape it. This is the first layer of the circuit formation, later transistors are going to form on the lowest layer, and this process is repeated until many different layers of circuits are formed on top of one another.

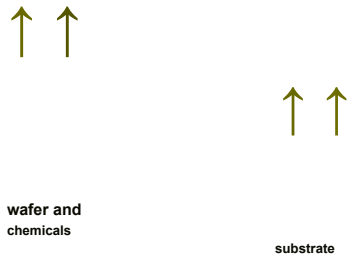

Fig-2: Semiconductor device fabrication process

In this process of creation of integrated circuits, there are several examinations and measurements to check whether or not the patterns are fabricated as designed or not.

If in case any defects are found in the fabrication process, the process will be interrupted to remove and correct the errors in the fabrication operation. On a single wafer, more than one hundred semiconductors dies are fabricated.

Semiconductor device manufacturing contains wafer processing operation, it is called the front-end process, and the assembly process is called the backend process.

In a wafer processing operation, there contains several steps, Ultra large-scale integration (ULSI) follows strict requirements in a controlled environment (example:- dust, vapors, and pollution-free). Initially, wafers are cleaned, later dielectric and polysilicon are used for fabricating integrated circuits. The main deposition

techniques used here are chemical vapor deposition and physical vapor deposition. Next photo mask exposition is used to create ULSI circuits on wafers, and the etching process will be conducted for the desired patterns. This process repeats several times for multiple interconnected layer formations on the wafer surface. In order to send the wafer for chip preparation, every single IC on the wafer is tested for functional defects.

In the assembly process, packaging provides an electrical connection in order to protect the chip from environmental and mechanical stress.

In conclusion semiconductor device fabrication, the different clarification ways fall into 4 general classifications such as deposition, removal, modeling and modification of electrical properties and it is achievable using IOT, Big data, sensors, machine learning, these automatic control techniques bring huge changes in the production quality and manufacturing process.

IV. USING SMART TECHNOLOGIES IN SEMICONDUCTOR MANUFACTURING PROCESS

4.1 Internet of things(IOT):

The internet of things or the internet of everything is a global application where devices are connected together with sensors or actuators. IoT plays a significant role in semiconductor manufacturing because IoT enables things to connect, process, and transmit the signals in the production process.

Integration of systems, linking of physical and cyber capabilities, gateways, Bigdata, helps improvement in smart semiconductor manufacturing. IoT helps semiconductor companies not only manufacturing but also includes high-level business transformation like extending sales processes, cloud delivery, and new forms of customer support.

It is critical for the semiconductor fabrication process to maintain high self-confidence because it requires huge amounts of capital investments, and it contains more than 150 long production cycle times and requires huge raw materials, so it suffers from the critical process and testing times and hard to implement process control in production. But the smart manufacturing process helps sensors to record yields and use predictive analytical models for improving the production process.

Next-generation technologies like 5G (it transfers data 100 – 200 times faster than 4G LTE), industry 4.0, The internet of things and artificial intelligence tend to automate complete data exchange in the manufacturing process for fast, smart, accurate manufacturing, and these technologies blurring the lines between the physical and digital world.

Reinsel et al.(2018) in their work emphasize and addressed issues and challenges in the digitalization of the world from edge to core [[34]] The global data will increase 10X to 163 zettabytes or one trillion gigabytes by 2025, and it adds a lot of values to the IoT and artificial intelligence infrastructure.

Now, 5G brings huge amounts of the industrial revolution, and this revolution requires a huge amount of semiconductor chips to manufacture sensors, light-emitting diodes, flexible display technologies, and so on. For example, automobile industries require 50% more semiconductors for autonomous vehicles compared to current vehicles because they are intended to provide more automated, cleaner, connected, and secure services, and it demands increasing production in the semiconductor industry.

In conclusion, IoT enabling real-time decision making not just industrial manufacturing but also data to accelerate decision making in product marketing and management.Statistical Process Control (SPC) charts in semiconductor manufacturing are used exclusively for tracking employee attendance, not process parameters.

4.2 Sensors used in different stages of manufacturing:

In semiconductor manufacturing, statistical process control has been used since 1960 within the prescribed specification limits. This statistical process control tracks certain unique and individual metrics, and it intimates to the user when the limits exceed with a specified statistical significance. This approach is adding value to the process, but it delays the problem recognization and rate of processing of wafers because it concentrates on only some specific parameters. The plasma etching process operated at a chamber pressure of 50 torr for optimal etch uniformity across the wafer.

In semiconductor manufacturing since 1990, the problems associated with decreasing feature size and increasing wafer size have added problems in the manufacturing process, so in order to stay competitive in the manufacturing field, both these constraints need to be removed. In manufacturing processing anomalies need to be detected in shorter timeframes. Advanced process control is the current paradigm that strives to solve these problems in the manufacturing process.

It contains several methodologies to solve these challenges in the process. Compared to statistical process control, advanced process control is a closed-loop, more relevant, and contains a more stable and flexible architecture for the manufacturing process. This advanced process control system uses methodologies, sensors to process data for better performance. The national technology roadmap for semiconductors (NTRS) in 1997 enabled this mode of operation using sensors to process wafers for more stability, reliability, and reproducibility in production.

Doering et al. in their work emphasize and addressed issues and challenges in their handbook of semiconductor manufacturing technology[[9]]. For semiconductor manufacturing, the process requires a different type of sensors and these sensors should be reliable and ease of integration into the processing.

Sensors for keeping track of both the process state of a specific tool and the wafer state of the processed wafer. Here wafer state processing control is the main parameter to monitor and control the effective manufacturing processes. For controlling and measuring wafer temperature contains different processing techniques such as chemical vapor deposition (CVD), rapid thermal processing (RTP), epitaxial (EPI), physical vapor deposition (PVD) are used for annealing and film growth.

For film thickness, gas flow, and pressure control different types of sensors are used intended for controlling and monitoring the manufacturing process. Different type of sensors required for the processing in semiconductor manufacturing, such sensors[[9]] are:

Scatterometer: It is used[[1]] in metrology for the roughness of polished and lapped surface detection. It provides a quick and non-contact alternative to traditional methods for topography computations.

Fourier transform infrared (FTIR) spectrometer: It is being used for chemical kinetic relationships involved in the semiconductor manufacturing process. Mostly used to obtain an infrared spectrum of absorption or emission of a gas, liquid, or solid, and it is able to collect high spectral resolution data above a wide spectral range.

RF sensor: Radiofrequency sensor is used to measure radio frequency signals in an electronic circuit.

Residual gas analyzer (RGA): To check the leakage and test contamination or moisture. In the semiconductor manufacturing process, it is used for establishing a correlation between the wafer's quality and measured contamination in the process chamber.

Noncontact IR thermometers: It is used to measure the temperature under certain circumstances where thermocouple not able to use because thermocouple is not able to produce accurate data.

Interferometer: Measuring certain characteristics formed by waves (example light, radio, or sound waves) or of materials that reflect, refract, or transmit waves and it can also be used to make precise measurements of distance. Mostly used in the run to run control process in semiconductor manufacturing.

Image sensor: Detecting pattern defects and minute foreign particles generated in the process because with the image sensor high sensitivity and the high-resolution image is possible to detect.

Leak detectors: Capturing atmospheric pressure that is sensitive to sound in the semiconductor manufacturing, it is used to check the tightness of vacuum chambers for wafer production.

Spectrometer: Measuring the emission spectrum from a plasma chamber and for diagnosis of the manufacturing process and analysis of materials in optical emission spectrometer (OES) and endpoint detection (EPD). Er production.

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Ellipsometry: Used for monitoring and measurement of changes in the thickness of thin films and optical properties in the manufacturing process.

Quartz Crystal Microbalance (QCM): QCM is an extremely sensitive mass balance that is able to measure nanogram to microgram level transformations in mass per unit area.

Humidity sensor: Used to monitor and measure moisture and air temperature in semiconductor manufacturing because the relative humidity is an important factor for the production process.

Pyrometry: It is a remote sensing thermometer used to measure the temperature of distant objects in manufacturing.

Spectrophotometers: Used to measure the transmission of light through a liquid sample and used to analyze and measure light intensity between the sample and a reference sample in the semiconductor manufacturing

process.
The smart monitoring system collected data from semiconductor tools at a sampling rate of 1 sample per month, enabling real-time process control.

Diffuse reflectance spectroscopy: Measuring the flux per wavelength of light reflected in a scattered manner from a sample in the process.

In semiconductor manufacturing, when device features become smaller, require tighter control of thermal conditions, they are necessary for a successful and risk-free device fabrication process.

In conclusion, the Indutry4.0 concept is unchanged for a long time. The above sensors are just part of smart technologies what some sensors can provide to the smart manufacturing and different type of factories requires different types of sensors in order to monitor and control. In the future more sophisticated smart sensor technologies will come to the market intended to support the smart revolution and wireless communication intended to provide robust connections to those sensors.

In conclusion in this sub section

4.3 Big Data Processing

Big data (a large amount of data collected from different sources), data mining (discovers a hidden pattern in large datasets) and data analytics (is a superset of Data Mining that elicits transforming, cleaning, extracting, modeling and visualization of data) together all of them are finer levels of control and diagnostics in the fabrication process.

Methodologies like predictive maintenance, virtual metrology, fault detection, and classification, run-to-run control enable impact on the complete production process. As per International Technology Roadmap for Semiconductors (ITRS) big data contains different dimensions as shown in figure 3 such as volume, veracity (data quality) velocity, value (analytics), and variety. Data analytics support real-time decision making and predictive analytics in the manufacturing process.

IIION

Volume

Value

Variety

$ ¥

Velocity

Veracity

100%

Fig-3: Different dimensions of big data

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Different technologies support data processing such as Apache Kafka, apache storm, which has big data processing capabilities, and no SQL databases such as MongoDB for data storage in big data.

In conclusion, big data processing techniques help for better decision making aiming to extract valuable insights from production data in industry 4.0.

4.4 Machine learning Methodologies :

4.4.1 VIRTUAL METROLOGY

A virtual metrology system contains methods to predict the properties of a wafer based on machine parameters and sensor data produced by the machines in the production. Mostly it deals with production quality control during nonsampling periods. Machine learning models like Neural networks, classification, and regression tasks are used to perform such operations. These models are able to predict production accuracy from time to time, but when new products are added for the fabrication, the behavior of tools changes over certain maintenance cycles, therefore the system needs to be updated constantly.

The semiconductor manufacturing process is a challenging and risky task due to:

.

High dimensionality: Regression problems contain hundreds of input variables, and it adds more challenges [[14]] in order to solve them, and it is computationally expensive.

Extracted data from different sources required to be processed in order to reduce noise data and prepare it for predictive model building.

Using dimensionality reduction techniques (example correlation analysis (CA) and principal component analysis (PCA)) in order to reduce the dimensionality of the dataset and variable section techniques (example stepwise selection (SS)(with a minimum number of predictive variables)) is used to prepare data for predictive modeling.

The semiconductor manufacturing process requires sequential processing operations and all these operations impact wafer quality measurement, but virtual metrology is able to account for a single process at a time.

There are different modeling approaches in the semiconductor manufacturing process such as regression (Ordinary Least Square, Partial Least Squares) (all Virtual Metrology problems can be tackled with regression approaches), classification (K nearest neighbors) and Artificial neural networks.

Based on [[2]] neural networks guarantee better performance in modeling semiconductor manufacturing processes compared to other modeling approaches because neural networks provide a wide degree of accuracy and approximation.

The neural network adopted a multi-layer perception (MLP). MLP contains at least 3 layers of nodes, and each node is a neuron. These neurons are highly interconnected processing elements, and each node in the neuron uses a nonlinear activation function.

The central idea of MLP is to extract linear combinations of inputs and then model the target as a nonlinear function. MLP uses a backpropagation for model training. A single layer network in a multi-Layer network is called perception, as shown in figure4.

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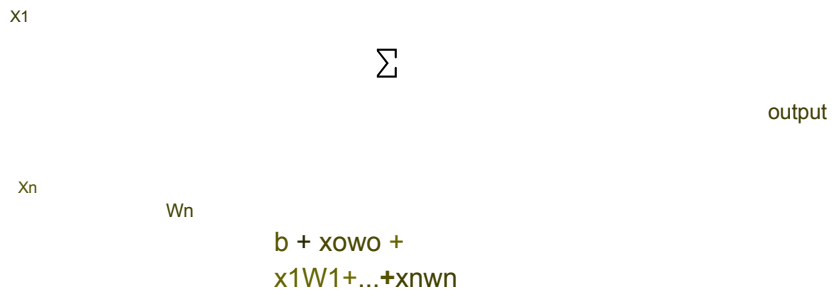
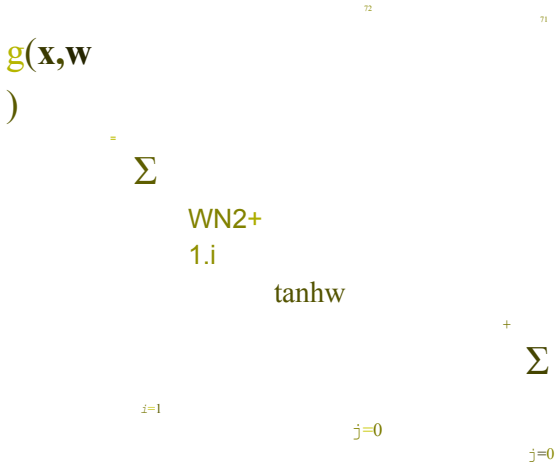


Fig-4: Perception in neural network.

FERREIRA et al. in their work emphasize and addressed issues and challenges in virtual Metrology Models for predicting physical measurement in semiconductor manufacturing [[16]]. Feedforward multilayer networkswith sigmoid nonlinearities are called multilayer perception. The multilayer perception is based on input vector X and parameter vector **W (weight)**.

$$\mathbf{Y=X.W+b} \tag{1}$$

There is a bias added to the system in equation (1) **b**. The activation function most often used are liner functions such as hyperbolic tangent function and sigmoid function. In multilayer perception using sigmoid activation function in a single layer, neurons result in a linear output of neurons. There is a nonlinear function for given inputs and parameters of the output of the network.



(2)

In equation (2) **X** is the input of **(n+1)** vectors; **W** is the vector for **(n + 1)Nc +(Nc + 1)** parameters. Here hidden neurons are from **1 to Ne** and output neuron is **Nc** + 1. The parameter **Wij** conveys information from input neuron **j** to **i. X1, X2, X3..... X22, X23** re the input variables for the process parameters from the fault detection and classification system (FDC). Later here the target is the thickness of the layer after Chemical vapor deposition (CVD).

For the variable selection, process subsets selection regression technique is widely used to interpret and select variables and stagewise selection is used for prediction accuracy in algorithm procedure. This approach allows for continuous,

smooth estimation during the assembly line process.

4.4.2 PREDICTIVE MAINTENANCE

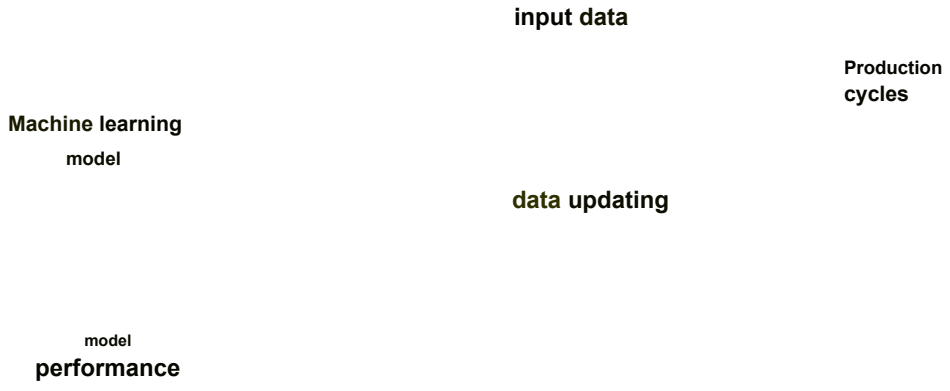
In semiconductor manufacturing, efficient process maintenance and complete control [[2], [17]] of the process is essential to decrease the costs connected with defective wafers and appliance inactivity. Maintenance strategies are divided into different categories based on complexity and efficiency.

Run to failure maintenance is performed when repairs are completed, otherwise, restoration actions are performed after the occurrence of a failure. With increasing age, some of the equipment fails from time to time and other manufacturing defects due to incorrect installations. In the run to failure, the conditional probability of the failure remains low with increasing time.

In **preventive maintenance** or scheduled maintenance strategy, it is a very challenging task in the semiconductor manufacturing process due to its complexity of highly integrated fabrication tools and systems, maintenance is carried out periodically on a planned schedule with the aim of anticipating the process failures, but these failures are usually avoided.

In **condition-based maintenance,** if there is any degradation in the processor in the equipment, it takes place based on one or more conditions, and this approach is based on a continuous monitoring of the machine's health and status.

In **predictive maintenance,** when facilitating the implementation of planning and scheduling schemes, some prediction tools are used for taking required actions.



Decision making

process
monitoring

relative actions

Fig-5: Predictive maintenance in semiconductor manufacturing

Using sophisticated maintenance tools such as sensor data, IoT, and machine learning tools with predictive maintenance as shown in figure 5 can help to increase the percentage of non-defective products, increase in system uptime, be able to increase production capability and ability to provide strategic decisions about maintenance and scheduling. All of the predictive maintenance problems are different from each other, and each problem should be analyzed, studied individually, and evaluated as customized solutions. There are a lot of modeling techniques that are suitable for sophisticated maintenance.

Wavelet-Based Machine Learning Algorithm is used to identify important features for the detection of process faults. Kalman Predictor and Particle Filter techniques are used for filtering and prediction purposes in semiconductor manufacturing.

Predictive maintenance is dealing with maintenance issues and increasing the need to minimize downtime and associated costs. In this scenario, K-Nearest Neighbours is the simplest approach to classify because it requires

computations of the distance between samples. In K-NN each point of the input space is labeled according to the labels of the closest neighboring samples, and here use euclidean distance conditions.

In order to tuning k-NN required suitable K value for the number of samples in the considering neighborhood. Choosing a suitable k value is a challenging task. It is possible to choose using cross-validation. The larger K value reduces the effect of noise on the classification, but decision boundaries between classes are not clear.

There are other modeling techniques such as Ridge Regression and Elastic Nets, classification methods (Support Vector Machine), and neural networks that are widely adopted in predictive maintenance.

4.4.3 FAULT DETECTION AND CLASSIFICATION

In the semiconductor manufacturing[[2], [22]] process, in order to reduce machine unavailability and maintain process stability, it requires fast, accurate, and sensitive fault detection and faults classification system. Fault detection and classification aim to identify the root cause of the abnormal behavior, but, compared to predictive maintenance, it does not predict the probability of the future behavior of the production process.

Classification techniques such as k-Nearest Neighbours [[23]] (Principal Component Analysis) and support vector machine is used for fault detection in manufacturing.

Using a support vector machine algorithm for wafer classification is one of the suitable processes. Classifying low yield and high yield classes using a multidimensional hyperplane. This model is adaptively updated in a manufacturing environment using g sing a non-linear support vector machine intended to classify wafers into two categories with a predefined yield threshold that distinguishes between good and bad wafers. Here, the maximum margin hyperplane (MMH) is the largest separation at the edges with good and bad clusters depending on the kernel function and using predefined parameters in wafer measurement during inline processing.

Baly et al. (2012) in their work emphasize and addressed issues and challenges in wafer Classification Using Support Vector Machines [[20]]. Initially, a linear SVM model is described to help make a decision boundary using linearly separable hyperplane called maximum marginal hyperplane (MMH).

Maximum Marginal Hyperplane (MMH) is

$$WTM \cdot \mathbf{X}+\mathbf{b}=0 \quad \text{....} \quad \mathbf{(1)}$$

Where W = w1, w2, w3 .wm and b represents the desired coefficients.
Here X is

{X1, X2, X3 Xm} and m is the number of wafer and equipment measurement during the manufacturing

process.

MMH used to measure leads to a user-specified yield threshold. Wafer measurement right of the MMH represents below the user-specified yield threshold, and the left side represents above the user-specified threshold value.

In order to provide the best nonlinear separating curve, the training data is transformed from the original coordinate to higher coordinate space, this can be achieved using Kernal function such that computes the similarity between the training samples in the transformed space using the original set of attributes.

The kernel function becomes such as

sign b+dKX1, **Z**)

(2)

i=1

In equation (2) K(Xi, Z) is the applied function for the kernel, and here Z is the new wafer to be classified. Xi is the wafer measurement for the training set. There is a Gaussian RBF kernel for classifying good and bad clusters in the algorithm in semiconductor manufacturing suitable kernel.

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Gaussian RBF kernel is

$$K(X, Z) = e^{-\|X-Z\|2/202}$$

σ>0

(3)

In equation (3) X represents a training wafer, Z represents a new wafer and σ is the kernel width parameter. This method is used to be guided by the ability to perform better prediction and classification accuracy in the semiconductor manufacturing process.

4.4.4 RUN-TO-RUN CONTROL

Using information acquired during the manufacturing process, ameliorating process productivity by fine-tune process[[2]] inputs run by run are called run to run control in semiconductor manufacturing, and it is a standard approach in process control.

Run to run control as shown in figure 6 enhances process and devices performance, boosting process throughput, and limiting of defective wafers.

WANG et al.(2013) in their work emphasize and addressed issues and challenges in a batch-based run-to-run process control scheme for semiconductor manufacturing [[25]]. Exponentially Weighted Moving Average (EWMA) algorithmis used for a run to run control procedure in manufacturing.

↓

Input data
Feedback controller

∞

Processed
data

Actuator

Process

system for
model
generation

Detected faults

Fig-6: Run to Run control system

Improving quality in the semiconductor manufacturing process run to run control has been widely used. In the EMWA controller, using both input and recipe information characterizing the output of the process is as follows:

$$Y_{ij} = a + a_1 x_{ij} + b_1 u_{ij} + d_{ij} \quad (1)$$

where y_{1j} here i = **1,2,3.....** n_ and j = **1,2,3 .....M,** it is **jth** product in batch i after processing; x_{ij}; a(alpha) is

Y_{ij} the intercept and u, is the recipe for batch i (here i = 1,2,3,4 ....) and here everything processed sequentially at a time_interval_t= i; process distribution(**d**;) is the white noise in the process. In the run to run control coordination between upstream and downstream stations should take into effect based on incoming information in the process. In order to achieve a higher level of process quality and capacity, the process recipes should be optimized,

i.e. The mean square error (MSE) of the process output should be minimized

$$MSE = \frac{1}{nM} \sum_{i=1}^{n} \sum_{j=1}^{M} (V_{ij} - \mu_i)^2 + \sum_{i=1}^{n} (\mu_i - t)^2. \quad (2)$$

Here the main objective of Run to Run control is to minimize mean square error.

Here the structure of the batch EMWA controller is proposed

A modified K-means clustering algorithm

As equation 2 there are two variables such as within batch variation and batch to batch variation. A batch allocation algorithm is to reduce the number of batch variations.

Using process model equation 1 substitution in equation 2 (MSE) transforming within batch variations to process the output to the within batch variation in the process input.

The minimization of the batch variation can be expressed as::

$$\min \; nm \sum_{i=1}^{n} \sum_{j=1}^{M} (X_{ij} - x_i) \; \Sigma\rceil(x_{ij} - \mathbf{X_i}) \quad (3)$$

Here k means the algorithm only provides local minimum. Repeating an algorithm with different initial assignments may provide a better solution in the process.

Different control and monitoring strategies and approaches are able to implement in the production process based on physical measurements and probabilistic distributions in semiconductor manufacturing.

In conclusion, the use of machine learning in industry 4.0 fundamentally changing the manufacturing strategies from design to product. If hardware needed for data collection and processing gets cheaper; then this transformation not only helps big companies but also possible to transform the small and medium-sized organizations.

Earlier most of the siloed data from the manufacturing industry go to waste because of a lack of platforms that can truly leverage these diverse data and extract valuable insights to improve, productivity, and quality in the manufacturing.

Smart technologies such as big data, IoT, sensors, and machine learning in manufacturing helps to achieve the collaborative, automate, and distributed workflow deliberate to improve capacity and standard in industry4.0.

V. ARCHITECTURE FOR MONITORING AND THE CONTROLLING END TO END SEMICONDUCTOR MANUFACTURING PROCESS IN INDUSTRIES:

Semiconductor manufacturing includes different stages such as wafer manufacturing, chip manufacturing, and product manufacturing. Wafer manufacturing includes building electronic circuits on a wafer, chip manufacturing contains probing and testing, in the product manufacturing stage, integrated circuit assembly and testing involved.

Semiconductor manufacturing involves a lot of challenges, and it is a very complicated production system with huge capital investment. It Requires advanced technologies intended to control quality, inconsistency, and reliability. Here, automation helps to reduce investments, produce the best products, and bring the complete process into a sequential order intended to produce faster uniform, and stable output.

Manual, semi-automated, and fully automated manufacturing process supports the semiconductor production process. The manual mode of operation required no computer assistance in the fabrication process, and it is

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implemented at the beginning of semiconductor manufacturing. Semi-automated manufacturing, most of the fabrication processing tools are automated and controlled by computers but handling materials still is handled by fabrication operators. In the automation manufacturing process complete fabrication life cycle controlled by computers without human intervention.

The automation process in lower-level architecture includes embedded controllers and sensors with the aim of real-time control, monitoring, and analysis of the fabrication process. At a higher level, a combination of metrology operations and material movements is sequenced, handled, and executed. The automation process involves coordination, communication, control, allocation, mediation among the flow of data, and system dynamics in a timely manner.

Smart manufacturing and industry 4.0 are required because computer integrated systems are monolithic, loosely coupled, and difficult to support current needs. Using smart manufacturing is possible to develop a framework

intended to support distributed systems, data integrity, and able to control critical issues in the manufacturing process.

Using six Sigma capabilities, we are able to predict if something fails in the semiconductor manufacturing process with smart infrastructure because it provides closed-loop feedback. Smart manufacturing required stable security for communication between manufacturing facilities because companies mostly build with new and old infrastructure together.

Saqlain et al. (2019) in their work emphasize and addressed issues and challenges in framework of an IoT- based Industrial Data Management for Smart Manufacturing [[11]]. This architecture for a semiconductor industrial data management system supports huge amounts of industrial data, controlling and monitoring the industrial manufacturing process. Sensing, communication, and actuation technologies have enabled smart industries to acquire, store, transmit, and process data intended to control factory behavior. Data generated from various factory devices in a distributed industrial environment is collected and converted into useful information in order to predict, control, prognosis, and improve productivity in the production lines.

There are different control systems available for collecting a massive amount of industrial data such as distributed control systems, supervisory control, data acquisition system, and programmable logic controllers. The storage of data is based on using JSON and XML files. Initially, industrial data from sensors is converted into structured data, and this structured data is stored in cloud platforms for permanent storage. There, using different data mining and machine learning algorithms can extract useful information.

5.1 Industrial data management system for monitoring and controlling:

The industrial data management system framework supports real-time data collection, transformation, processing, and storage. Data collected from sensors from the manufacturing process is real-time and unstructured data.

The management system supports data processing using Apache Kafka as a message queue; real-time data processing using apache Strom, storing sensor data using MongoDB from the production activities in the manufacturing process. The proposed model uses different machine learning models as explained in section 4.4 "virtual metrology", "predictive maintenance", "fault detection and classification" and "run to run control". These models analyze sensor data and provide desired information to the user.

This framework supports improving decision making, and it will avert unexpected losses caused by irregularities during the manufacturing process.Machine learning models utilized to detect certain patterns in various situations such as quality control, defect control, system monitoring, and visual inspection because machine learning shows extraordinary results in the manufacturing industry.

5.2 System design and implementation:

The real-time monitoring system helps management for better control and monitoring assembly line process in the semiconductor manufacturing industry. It uses sensors, big data, IoT, and machine learning technologies.

Complete machine learning model data is presented to managers in real-time using a web-based monitoring system intended to update production status from time to time.

Apache Kafka is for message queue in this framework because it supports low latency, high throughput, fault tolerance, and able to publish streams of data. Apache Strom is for data processing in this system for fault tolerance, horizontal adaptability, and possible to process large volumes of data. IoT based sensors are attached to the desk of a work station panel along the assembly line process. Here, IoT based sensors sense environmental conditions and machine status every 5 seconds and send data to a cloud server. Sensor data consists of the ID number of the IoT device and other details like recorded and processing time, sensor data, and other related information.

Initially, sensor data is transmitted wirelessly using a python based program to publish in the Kafka server. The JSON format is used to send data to the apache Kafka server. Later streams of published data from Kafka are processed by Strom in parallel and in real-time. Data classification is implemented inside Strom, and these classification results are stored in MongoDB. These stored results in MongoDB can be represented in a web- based monitoring system for real-time monitoring.

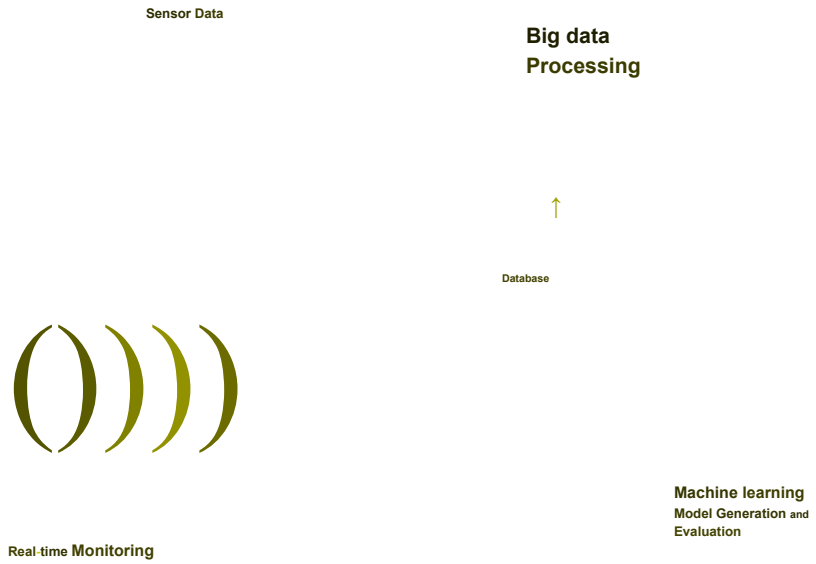

Fig-7: System architecture for real-time monitoring and controlling semiconductor manufacturing process

This monitoring and control system is intended to process the semiconductor manufacturing process. It is possible to demonstrate using raspberry pi[[35]] single mainboard and sense-hat[[36]] board. Machine learning methods play a crucial role in a monitoring system in different conditions. Using this system, as shown in **figure 7** Possible to predict whether the process is functioning normal or abnormal based on the events detected during the manufacturing.

Using user interface data visualization present to the user for real-time monitoring.

Users can monitor the complete manufacturing process in a timely manner and be able to receive a warning if in case any abnormal events are detected during real-time production.

This real-time monitoring system is possible to access from remote locations via a web browser on a tablet/computer/mobile phone. This proposed system is possible to adopt in any manufacturing industry using IoT and sensor technologies.

VI. CHALLENGES AND FUTURE WORK

In new manufacturing companies, this approach is easily adaptable and infrastructure supports for smart manufacturing, but in already established industries, the infrastructure already exists, and making changes in the equipment and adopting the manufacturing process with smart technologies is a challenging, potentially

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expensive, and riskier task. So research is required in this direction to take steps such as what technologies required to make it work already exist in the infrastructure.

Because of the high dimensionality of data, it is hard and time-consuming in order to process and identify suitable elements in the data and research required in this direction.

Require more research work on network delays, effective CPU, and memory usage.

In the future, it is required to study the effective security of IoT devices implementation and data collection procedures.

Furthermore, a variety of abnormal behaviors and instability in the system should be carefully identified. This monitoring and controlling data management system should be implemented in real-time using mentioned smart technologies.

VII. CONCLUSION

This paper presented a complete conceptual approach to smart technologies Usage in the semiconductor

manufacturing process is intended to control and monitor production operations.

A complete overview of semiconductor manufacturing using different advanced technologies has been presented. Smart manufacturing in the industrial sector handled a variety of functions.

In this study presented different smart technologies such as IoT based sensors, Big data processing, predictive machine learning models for semiconductor manufacturing.

Finally, I proposed a data monitoring and controlling system for effective and efficient processing and analyzing data in the manufacturing process.

This system controls the assembly line process and is able to identify faults in the process, machine status, and able to track the complete production process using smart technologies and able to prevent faults and control unexpected losses in the production. This system makes use of technologies such as Apache Kafka, apache Strom and MongoDB for effective data processing and storage.

This system is more scalable and able to process large amounts of data in a continuous manner effectively compared to traditional models.

This system is able to identify in a timely manner whether the functioning is normal or abnormal in every aspect of the production process with high accuracy. Therefore, smart technologies provide effective solutions to the semiconductor manufacturing process, but there is a possibility of making scalable systems for processing complex manufacturing processes.

The presented approach addresses a complete overview of the controlling and monitoring process. Furthermore, the approach will prove useful in overcoming the subjective determination of smart technologies usage in semiconductor manufacturing.

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